

This should help. But don't memorize without understanding :) You might forget or get confused! Also your formulas might change depending on the scenario, for example in some questions, one more bit called dirty bit is given as meta data in cache.

1. Line Number = Block Number % Total Number of Lines
2. Number of Blocks = Physical Memory Size / Block Size
3. Number of Lines in Cache = Cache Memory Size / Block Size
4. Physical Memory Bit Size / Address Size = $\log_2$ (Physical Memory Size)
5. Physical Memory Size = $2^{\text{address space size}}$
6. Index Field to determine Block = $\log_2$ (Total No of Blocks)
7. Index Field to determine Cache Line Number = $\log_2$ (Total No of Lines in Cache)
8. Block Offset = $\log_2$ (Block Size)
9. Block Size = $2^{\text{Block Offset}}$ [Block offset in byte if memory byte addressable, Block offset will be in word if memory is word addressable]
10. Tag Bit Size = Index Field for block - Index field for cache line
11. Physical Memory Address Space = Index Field for Block + Block Offset
12. Index Field for Block = Tag Bit + Index Field Size for Cache
13. Actual Size of Cache Memory = Number of Lines x (1 valid bit + Number of Tag bits + 1 Dirty bit [if given in question] + Size of cache [given in question or you have to find it, but you must convert it into bits]
14. Average Memory Access Time (AMAT) = Hit time + Miss rate $\times$ Miss penalty